BIOLOGY - THE STUDY OF LIFE

# Module map

- Module 01: Biology - The Study of Life
- Connected lab: lab-01_measurement-methods.md
- Use this deck as the visual path through the module's evidence and retrieval work.

## Module 01

Biology - The Study of Life

1. What counts as life
2. Science as a way of testing
3. Organization from cells to ecosystems
4. Evidence, variables, and measurement
5. How biological explanations improve

**Lab evidence**

lab-01_measurement-methods.md

**Teaching note:** Open by naming the module claim and pointing students to the lab as the evidence surface.

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# Learning objectives

- Distinguish living systems from nonliving examples using multiple characteristics.
- Separate hypotheses, predictions, observations, and conclusions in a scientific test.
- Place biological examples at the correct scale of organization.
- Identify variables and evidence in a simple investigation.
- Revise a biological explanation when new evidence appears.

**OBJECTIVE 1**
Distinguish living systems from nonliving examples using multiple characteristics.

**OBJECTIVE 2**
Separate hypotheses, predictions, observations, and conclusions in a scientific test.

**OBJECTIVE 3**
Place biological examples at the correct scale of organization.

**OBJECTIVE 4**
Identify variables and evidence in a simple investigation.

**OBJECTIVE 5**
Revise a biological explanation when new evidence appears.

**Teaching note:** Frame objectives as actions students should be able to perform, not facts to copy.

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# Topic sequence

- What counts as life: Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.
- Science as a way of testing: Scientific claims are tested with observations, comparisons, variables, and repeatable reasoning.
- Organization from cells to ecosystems: Biological systems are organized across scales from molecules to biosphere.
- Evidence, variables, and measurement: Data become useful when measurements are tied to a clear question and source of error.
- How biological explanations improve: Strong explanations connect evidence to a claim and stay open to revision.

01

**What counts as life**

Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.

02

**Science as a way of testing**

Scientific claims are tested with observations, comparisons, variables, and repeatable reasoning.

03

**Organization from cells to ecosystems**

Biological systems are organized across scales from molecules to biosphere.

04

**Evidence, variables, and measurement**

Data become useful when measurements are tied to a clear question and source of error.

05

**How biological explanations improve**

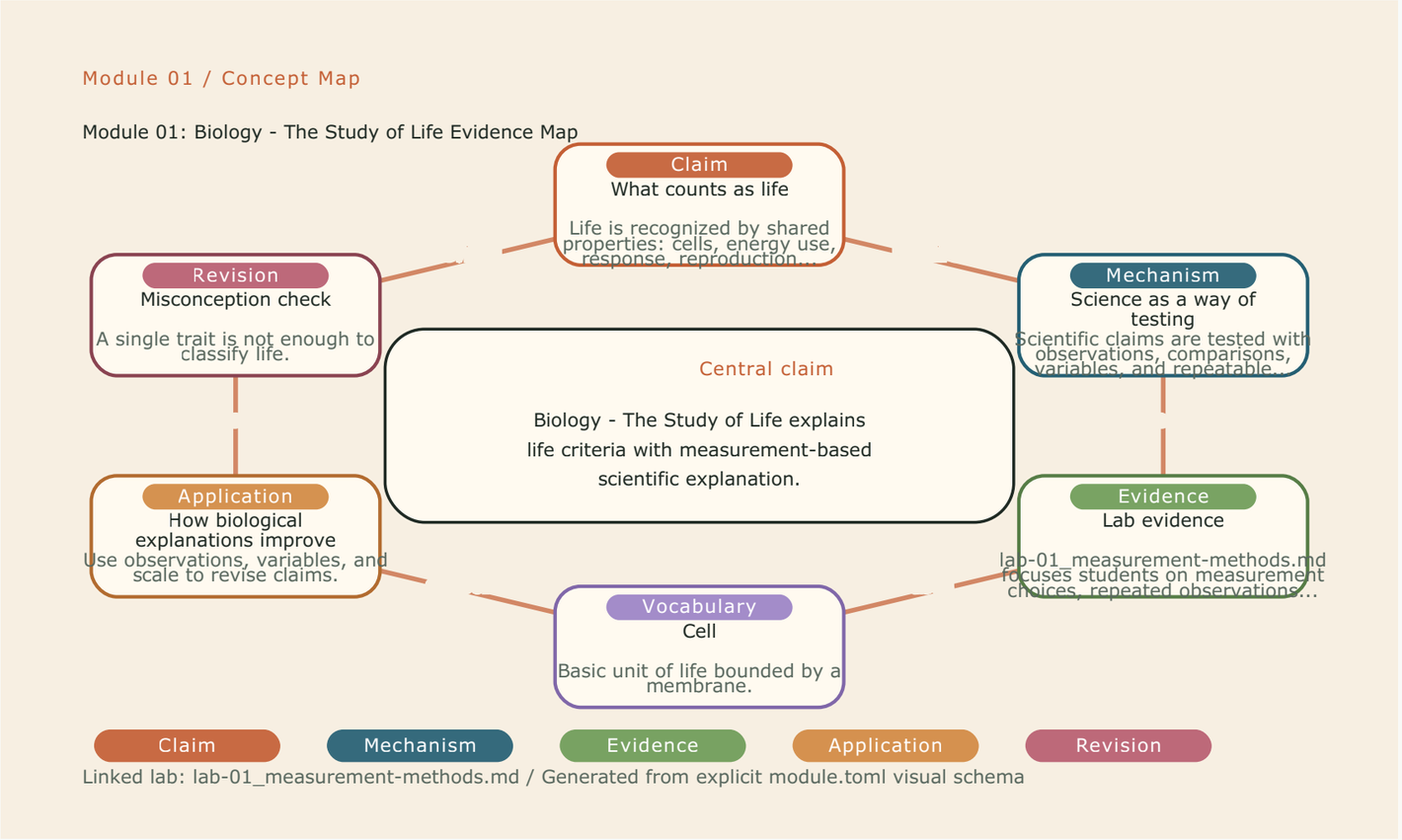
Strong explanations connect evidence to a claim and stay open to revision.

**Teaching note:** Show the module as an ordered explanation so students can locate each new idea.

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# Module 01: Biology - The Study of Life Evidence Map

- Module focus: Biology - The Study of Life
- Central claim: Biology - The Study of Life explains life criteria with measurement-based scientific explanation.
- Key nodes to connect: What counts as life, Science as a way of testing, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-01_measurement-methods.md supplies an evidence surface for the map.

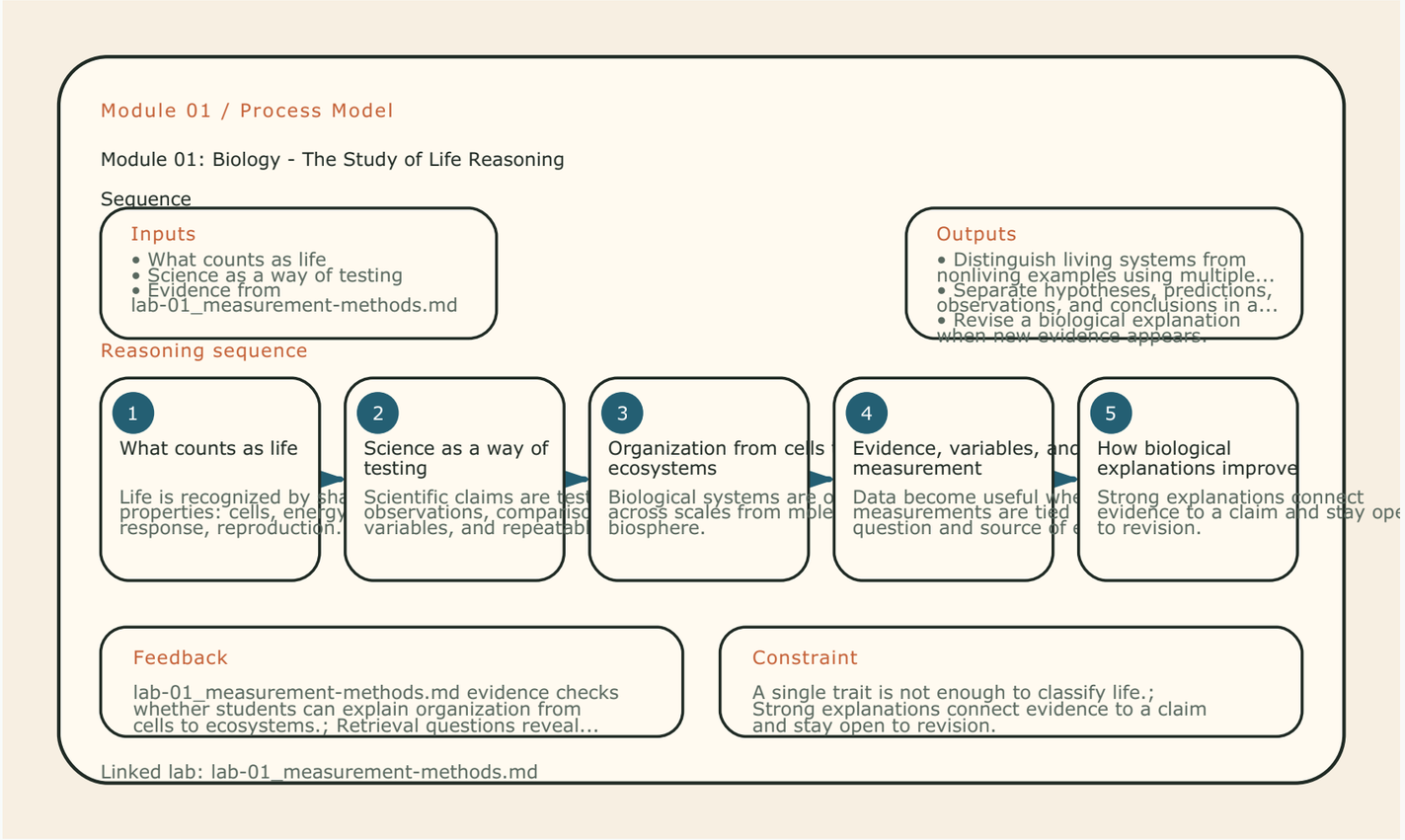



**Teaching note:** Ask students to explain one edge in their own words before moving on.

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# Module 01: Biology - The Study of Life Reasoning Sequence

- Module focus: Biology - The Study of Life
- Trace the model: What counts as life -> Science as a way of testing -> Organization from cells to ecosystems -> Evidence, variables, and measurement
- Inputs become outputs: What counts as life, Science as a way of testing -> Distinguish living systems from nonliving examples using multiple characteristics., Separate hypotheses, predictions, observations, and conclusions in a scientific test.
- Feedback check: lab-01_measurement-methods.md evidence checks whether students can explain organization from cells to ecosystems.
- Lab connection: lab-01_measurement-methods.md gives students a process to observe or test.




**Teaching note:** Emphasize where the process can fail, slow down, or be revised by evidence.

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# Terms as evidence handles

- Cell: Basic unit of life bounded by a membrane.
- Homeostasis: Regulation that keeps internal conditions within a workable range.
- Metabolism: All chemical reactions that move matter and energy through life.
- Hypothesis: Testable explanation for an observed pattern.
- Variable: Factor that can change in an investigation.
- Evidence: Observation or measurement used to support or challenge a claim.

**CELL**
Basic unit of life bounded by a membrane.

**HOMEOSTASIS**
Regulation that keeps internal conditions within a workable range.

**METABOLISM**
All chemical reactions that move matter and energy through life.

**HYPOTHESIS**
Testable explanation for an observed pattern.

**VARIABLE**
Factor that can change in an investigation.

**EVIDENCE**
Observation or measurement used to support or challenge a claim.

**Teaching note:** Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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# Lab connection

- Lab file: lab-01_measurement-methods.md
- Lab evidence should test or illustrate: Organization from cells to ecosystems
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

**QUESTION**
What counts as life

**EVIDENCE**
lab-01_measurement-methods.md

**CLAIM**
Distinguish living systems from nonliving examples using multiple characteristics.

**Teaching note:** Make the lab connection explicit before students encounter the activity.

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# Contrast check

- Do not stop at: Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.
- Stronger explanation: Strong explanations connect evidence to a claim and stay open to revision.
- The goal is to move from a named idea to a testable biological explanation.

**SURFACE**

Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.

**DEEPER**

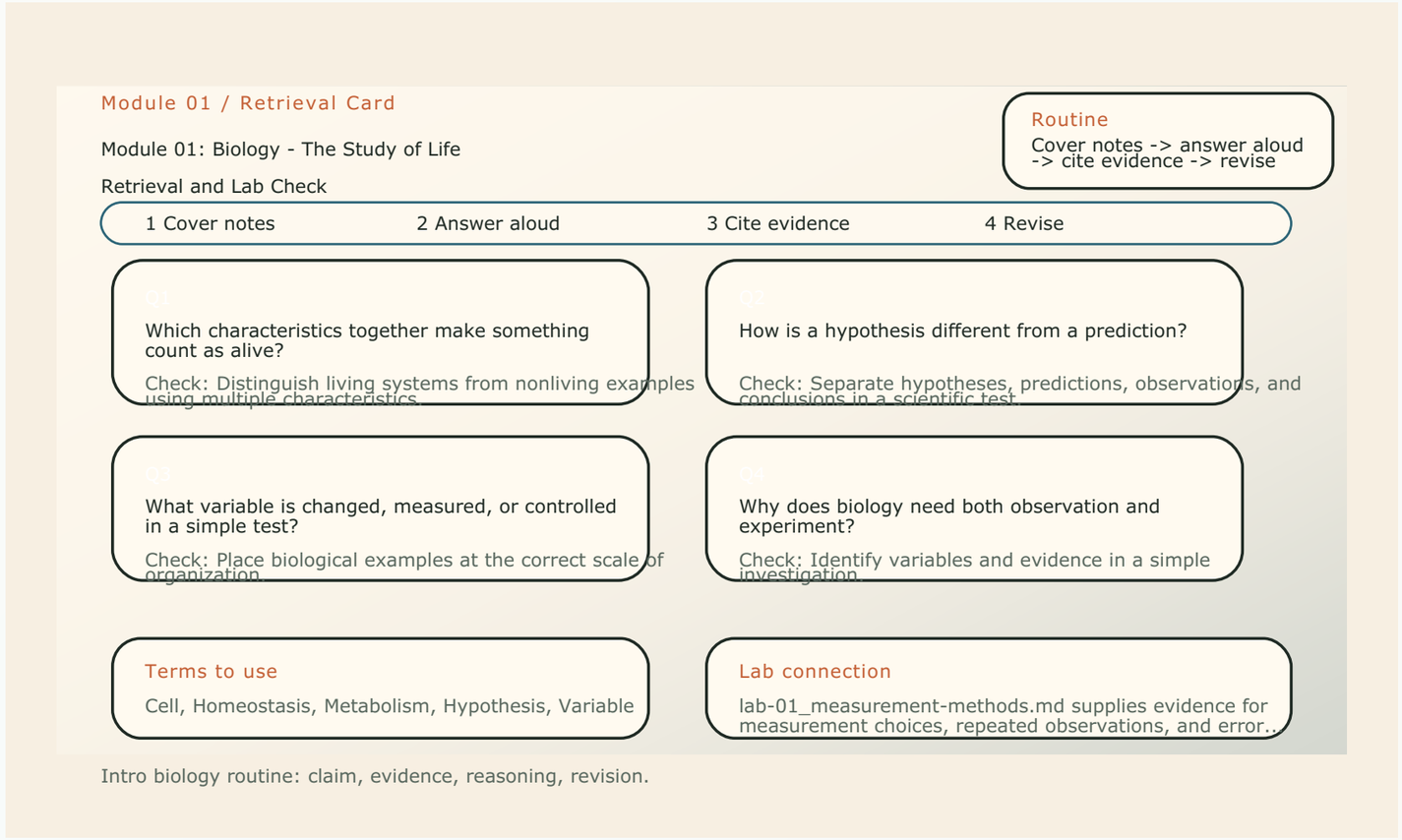
Strong explanations connect evidence to a claim and stay open to revision.

**Teaching note:** Use this slide to prevent shallow vocabulary-only learning.

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# Module 01: Biology - The Study of Life Retrieval and Lab Check

- Module focus: Biology - The Study of Life
- Retrieval prompt: Which characteristics together make something count as alive?
- Answer check: Distinguish living systems from nonliving examples using multiple characteristics.
- Required terms: Cell, Homeostasis, Metabolism
- Lab connection: lab-01_measurement-methods.md; lab-01_measurement-methods.md supplies evidence for measurement choices, repeated observations, and error sources.




**Teaching note:** Pause here. Students answer first without notes, then compare to the checks.

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# Practice quiz bridge

- 1. A student sees frost crystals slowly "grow" on a cold window overnight. Why don't biologists count this as life?
- 2. To test whether fertilizer speeds plant growth, a student changes the amount of fertilizer given to each plant. The amount of fertilizer is the:
- 3. Which list places levels of biological organization from smallest to largest?
- 4. A hypothesis survives many careful tests over many years. What should happen to it?

**QUESTION 1**

**Key: A**

A student sees frost crystals slowly "grow" on a cold window overnight. Why don't biologists count this as life?

**QUESTION 2**

**Key: B**

To test whether fertilizer speeds plant growth, a student changes the amount of fertilizer given to each plant. The amount of fertilizer is the:

**QUESTION 3**

**Key: C**

Which list places levels of biological organization from smallest to largest?

**QUESTION 4**

**Key: D**

A hypothesis survives many careful tests over many years. What should happen to it?

**Teaching note:** Use the quiz as formative feedback. Answer key: 1=A, 2=B, 3=C, 4=D

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# Synthesis and exit ticket

- Exit claim: explain what counts as life using evidence from lab-01_measurement-methods.md.
- Must include: Cell, Homeostasis, Metabolism.
- Revision prompt: what evidence would change your explanation?

**CLAIM**
What counts as life

**EVIDENCE**
lab-01_measurement-methods.md

**REVISION**
What would change your mind?

**Teaching note:** End with a claim-evidence-reasoning exit ticket connected to the same module data.